



BOARD OF INTERMEDIATE EDUCATION, KARACHI

HSC Computer Science · Science General & Humanities

Theory 75 marks + Practical 25 marks = 100 per part

COMPLETE CLASSROOM LECTURES · HSC PART-I

Computer Science XI

Original notes for Karachi Board (BIEK)

WHAT THESE NOTES COVER

- ▶ Sindh Curriculum 2019 Grade XI units 1–7 with board weightages
- ▶ Extra Paper-I topics: number systems, Boolean algebra, data security
- ▶ C++ programming, arrays, strings, structures, and lab journal list
- ▶ Paper pattern (75 theory + 25 practical) and a full model paper
- ▶ Definitions, comparison tables, programs, MCQs, short and long questions

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How to use these lectures

These notes are original classroom lectures written for students appearing in HSC Part-I Computer Science from the Board of Intermediate Education, Karachi (BIEK). They follow the public Sindh Curriculum for Computer Science, Grades XI-XII (2019), and also include the theory topics that still appear in BIEK Computer Science Paper-I: Boolean algebra, number systems, data security, and computer architecture.

- Read one unit, then attempt that unit's short questions without looking at the answers.
- Copy every C++ program into Turbo C++ / Dev-C++ / VS Code and run it.
- Memorise definitions in the blue boxes; board short questions are often one definition plus two examples.
- Use the end-of-book model paper under timed conditions (Section A 30 min, B 90 min, C 60 min).

Remember

These notes are not a photocopy or reconstruction of any commercial textbook. They are independently written teaching notes aligned to the public curriculum and BIEK paper pattern.

BIEK paper pattern (Computer Science I)

Each HSC part carries 100 marks: 75 theory + 25 practical. The theory paper is three hours. BIEK's published scheme for Science General and Humanities (Regular) is:

Section	Type	Time	Marks	What you must do
A	MCQs (complete syllabus)	30 min	29	Attempt all 29; 1 mark each
B	Short answers	90 min	30	15 questions given; attempt 10 (3 marks each)
C	Detailed answers	60 min	16	3 questions given; attempt 2 (8 marks each)

Board exam tip

Section C has no part-questions. Write a full 8-mark answer: definition, classification or diagram, and at least three explained points. Leave 5 minutes to underline key terms.

Syllabus map

Sindh Curriculum 2019 unit weightages for Grade XI, plus extra BIEK Paper-I topics still asked in Karachi board exams.

Unit	Title	Weight	Why it matters in BIEK
1	Overview of Computer System	15%	Definitions, hardware, software, AI/cloud
2	Computer Memory	10%	Bit-TB, RAM/ROM families, secondary storage
3	Inside the System Unit	15%	Motherboard, CPU, buses, ports, slots
4	Operating System	10%	Types, functions, process states, GUI
5	Programming using C++	20%	Data types, control, functions, overloading
6	Arrays, Strings, Structures	15%	1-D/2-D arrays, string.h, struct
7	Communication & Networks	15%	Media, topologies, OSI, TCP/IP

Unit	Title	Weight	Why it matters in BIEK
A	Number systems & codes	extra	Binary/hex, ASCII, BCD, EBCDIC
B	Boolean algebra & gates	extra	Laws, truth tables, De Morgan, K-map idea
C	Data security	extra	Virus, firewall, proxy, email safety

Unit 1 — Overview of the Computer System

Curriculum weight 15%. This unit builds the vocabulary used in every later chapter.

1.1 What is a computer system?

Definition — Computer

An electronic device that accepts data (input), processes it according to stored instructions, stores results, and produces information (output) with high speed and accuracy.

Definition — Computer system

The complete working combination of hardware, software, data, users (people), and procedures. A CPU chip alone is not a system; Windows alone is not a system.

IPO model: Input → Process → Output, with Storage feeding the process and holding results. The stored-program idea (von Neumann) means both data and instructions live in memory.

Block diagram (IPO)

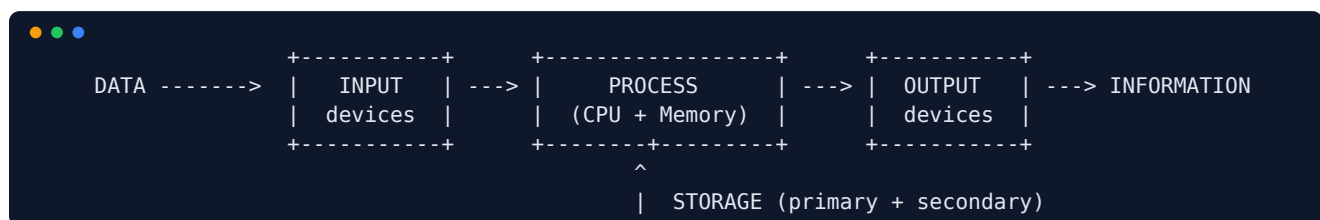


Fig. 1.1 Block diagram of a computer system

1.2 Characteristics

Feature	Meaning	Board example
Speed	Millions of instructions per second	Payroll of 5,000 employees in seconds
Accuracy	Errors come from humans, not the CPU logic	Wrong formula in a spreadsheet
Diligence	No tiredness or boredom	ATM works 24 hours
Storage	Huge data in tiny space	A 1 TB disk holds a library
Versatility	Same machine, many jobs	Accounts, games, design
Automation	Runs a stored program without step-by-step human control	Batch billing

Limitations you should mention in a long answer: no IQ or common sense, depends on a program, no feelings, and garbage-in/garbage-out (GIGO).

1.3 Generations of computers

Gen.	Period (approx.)	Main technology	Example / note
1st	1940s-1950s	Vacuum tubes	ENIAC, UNIVAC; huge, hot, slow

Gen.	Period (approx.)	Main technology	Example / note
2nd	1950s-1960s	Transistors	Smaller, more reliable
3rd	1960s-1970s	Integrated circuits (IC)	Minicomputers appear
4th	1970s-present	Microprocessor (VLSI)	IBM PC, laptops, smartphones
5th	Present-future	AI, parallel processing, ULSI	Expert systems, NLP, robots

1.4 Types of computers

By size / capacity

- Supercomputer — fastest; weather, nuclear research, climate models (examples: Cray-class machines).
- Mainframe — many simultaneous users; banks, airlines, tax systems.
- Minicomputer / midrange — department-level servers.
- Microcomputer / PC — desktop, laptop, tablet, smartphone.
- Workstation — high-end microcomputer for CAD, video, engineering.

By purpose / function

- General purpose — can run many kinds of programs (school PC, office laptop).
- Special purpose / dedicated — built for one job (ATM controller, washing-machine MCU, ABS in a car).
- Analog — continuous physical quantities (old laboratory instruments).
- Digital — discrete 0/1 data (almost all modern computers).
- Hybrid — analog plus digital (some hospital monitoring systems).

1.5 Hardware classification

Hardware is the physical part you can touch. Classify devices by the job they do.

Class	Job	Examples you must name in exams
Input	Send data into the system	Keyboard, mouse, scanner, mic, joystick, stylus, webcam, barcode reader, touch screen
Output	Present results	Monitor (CRT, LCD, LED, plasma), printer, plotter, speaker, projector, SGD
Processing	Transform data	CPU / microprocessor
Storage	Hold data	HDD, SSD, USB flash, SD/microSD, CD/DVD/Blu-ray
Communication	Move data between machines	NIC, modem, router, switch/hub, gateway, fax

Printers (favourite short question)

Type	How it prints	Quality / use
Dot-matrix (impact)	Pins strike an ink ribbon	Carbon copies, noisy, cheap running cost
Inkjet (non-impact)	Sprays liquid ink	Home photos, colour, clogging risk
Laser (non-impact)	Toner fused by heat	Offices, high speed, sharp text
Plotter	Pens or inkjet on large paper	Maps, CAD drawings
3D printer	Adds layers of plastic/resin	Prototypes (mention as modern extra)

Definition — Hard copy / Soft copy

Hard copy is printed on paper. Soft copy is displayed on a screen or stored as a file. A PDF on a USB is still a soft copy until it is printed.

1.6 Software at a glance

Software is the set of programs and associated data that tell hardware what to do. Unit 1 only needs the map; later units go deeper.

- System software — OS, device drivers, language translators, anti-malware.
- Application software — general purpose (Word, Excel, Chrome) and special purpose (hospital MIS, payroll).
- Utility software — compression, backup, disk manager, data recovery.

1.7 Cutting-edge topics required by the 2019 curriculum

Definition — Artificial Intelligence (AI)

The branch of computer science that tries to make machines perform tasks that normally need human intelligence: learning, reasoning, perception, and language.

Applications to quote: medical diagnosis support, face unlock, chatbots, self-driving research, fraud detection in banks, and recommendation systems. Mention ethics: bias, privacy, job change.

Definition — Cloud computing

Delivery of computing resources (servers, storage, databases, software) over the Internet on a pay-as-you-go model, instead of owning a local data centre.

- Advantages: lower capital cost, easy scaling, access from anywhere, professional backups.
- Examples: Google Drive, Microsoft 365, AWS, college email on Gmail.
- Caution: needs a reliable link; sensitive data needs a clear privacy policy.

Unit 1 — board practice

Short questions (3 marks)

Q. Differentiate hardware and software with two examples each.

A. Hardware is physical (keyboard, RAM). Software is a set of instructions (Windows, MS Word). Hardware is visible and wears out; software is intangible and can be copied.

Q. Why is a computer called a versatile machine?

A. The same hardware can run many programs — accounts today, video editing tomorrow — because the program, not the wiring, defines the job.

Long question (8 marks)

Q. Draw the block diagram of a computer system and explain input, process, output and storage.

A. Draw Fig. 1.1. Define each box with two device examples. End with IPO + stored program. That structure scores full marks.

MCQs

1. Which generation introduced the microprocessor?

- (A) 1st
- (B) 2nd
- (C) 3rd
- (D) 4th ✓

2. A plotter is mainly an

- (A) input device
- (B) output device ✓
- (C) storage device
- (D) OS

3. GIGO means

- (A) Good Input Good Output
- (B) Garbage In Garbage Out ✓
- (C) Gate In Gate Out
- (D) None

Unit 2 — Computer Memory

Curriculum weight 10%. Almost every Paper-I contains one memory MCQ and one short question.

2.1 Measurement units

Unit	Equals	Board memory aid
Bit	0 or 1	Smallest unit
Nibble	4 bits	Half a byte; one hex digit
Byte	8 bits	One ASCII character
KB	1024 bytes	About one short email
MB	1024 KB	A photo or MP3
GB	1024 MB	A film or a game
TB	1024 GB	A disk pack / backup drive

Common mistake

In board answers always write 1 KB = 1024 bytes, not 1000. Hard-disk marketers use 1000, which is SI, not the exam convention.

2.2 Primary / main memory

Definition — Primary memory

Semiconductor memory the CPU can address directly. It is fast and limited in size. Two families: RAM and ROM.

Point	RAM	ROM
Full form	Random Access Memory	Read Only Memory
Volatility	Volatile — contents vanish on power-off	Non-volatile
Read/write	Read and write	Normally read only
Use	Running programs and data	Firmware / BIOS
Speed	Very fast	Fast, but not a working area

RAM types

- SRAM (Static RAM) — flip-flops; no refresh; used as cache; expensive, small.
- DRAM (Dynamic RAM) — capacitors; needs refresh; cheap, large; used as main memory.

ROM types

Type	Can you write after manufacture?	How to erase
Mask ROM	No	Cannot
PROM	Once (by programmer)	Cannot
EPROM	Yes, after erase	Ultraviolet light through a window
EEPROM	Yes, electrically, byte-wise	Electric signal
Flash	Yes, in blocks	Electric; used in BIOS and USB

2.3 Cache and registers

Cache is a small, very fast SRAM sitting between CPU and RAM. L1 is on the core, L2/L3 are larger and slightly slower. Registers (accumulator, PC, IR, MAR, MDR, FLAGS) are the fastest storage of all — a handful of bytes inside the CPU.

Remember

Hierarchy from fastest/smallest to slowest/largest: registers → cache → RAM → SSD/HDD → optical/tape.

2.4 Secondary memory

- Hard disk drive (HDD) — magnetic platters; cheap capacity; moving parts.
- Solid state drive (SSD) — flash chips; faster, no spinning platter.
- USB flash / data traveller — portable EEPROM/flash.
- External HDD — backup and extra space.
- SD / microSD — phones, cameras.
- CD / DVD / Blu-ray / combo drive — optical; CD ≈ 700 MB, DVD ≈ 4.7 GB, Blu-ray 25 GB+ per layer.

Q. Differentiate volatile and non-volatile memory.

A. Volatile memory (RAM) loses data when power is removed. Non-volatile memory (ROM, HDD, flash) keeps data without power. A running document lives in RAM; Save writes it to secondary non-volatile storage.

Unit 3 — Inside the System Unit (Computer Architecture)

Curriculum weight 15%. Overlaps BIEK's old 'Computer Architecture' chapter — fetch-decode-execute is a classic 8-mark question.

3.1 The system unit

The system unit (CPU cabinet / chassis) houses the power supply, motherboard, storage drives, cooling, and ports. Form factors: desktop tower, SFF, all-in-one, laptop clamshell. Cooling uses heatsink + fan, and on some chips, liquid loops.

- Power supply (SMPS) converts 220 V AC to 12 V / 5 V / 3.3 V DC.
- Motherboard is the main printed circuit board.
- Hard drive / SSD and optical combo drive mount in bays or M.2 slots.
- Ports on the back/front panel connect peripherals.

3.2 Motherboard components

Part	Function
Microprocessor socket	Holds the CPU
Chipset (north/south or modern PCH)	Traffic manager between CPU, RAM, and I/O
RAM slots (DIMM)	Main memory modules
BIOS / UEFI chip	Firmware that starts the machine
CMOS battery	Keeps clock and some setup values when unplugged
Jumpers	Tiny switches for old configuration options
Expansion slots	PCI / PCIe cards: graphics, Wi-Fi, capture
System bus	Shared pathways for address, data, control
Ports	USB, HDMI, audio, LAN, legacy serial/parallel

3.3 Microprocessor

Definition — Microprocessor

An IC that contains the ALU, control unit, and registers of a CPU on a single chip. It fetches instructions from memory and executes them.

Internal architecture

- ALU — arithmetic (+ − × ÷) and logic (AND OR NOT XOR, compare).
- Control Unit — fetches instructions, decodes opcodes, issues control signals, manages timing.
- Registers — scratch storage; Program Counter holds the next instruction address; Instruction Register holds the current instruction.
- Clock — crystal pulses; 3.2 GHz means 3.2 billion ticks per second (not the same as 3.2 billion instructions, because of multi-cycle ops and IPC).
- Cache — on-chip SRAM.

External pins (idea)

Address pins send the memory address, data pins carry the bits, control pins carry READ/WRITE/MEM/IO, interrupt pins let devices request service, power pins supply Vcc/GND.

Specifications to quote: process technology (nm), generation/family, clock speed, word size (32/64-bit), bus width, cache size, number of cores, TDP.

3.4 Buses

Bus	Carries	Direction
Address bus	Location numbers	Unidirectional (CPU → memory/IO)
Data bus	Actual data and instructions	Bidirectional
Control bus	Read, write, interrupt, clock	Mostly CPU → others, some status back

Board exam tip

Width of the address bus decides how much memory can be addressed: n address lines → 2^n locations. A 32-bit address bus theoretically maps 4 GB.

3.5 Ports and expansion slots

Port / slot	Typical use
Serial (legacy RS-232)	Old mouse, industrial devices; bit by bit
Parallel (legacy)	Old printers; several bits at once
USB 2/3/C	Almost every peripheral; hot-pluggable
HDMI	Digital video + audio to monitor/TV
VGA	Older analog video
PCI	Older expansion cards
PCIe (modern; curriculum also lists EISA/VISA as historical)	Graphics and fast I/O

3.6 Fetch-Decode-Execute cycle

This is the heartbeat of the von Neumann machine. Write it as three (sometimes four) labelled steps.

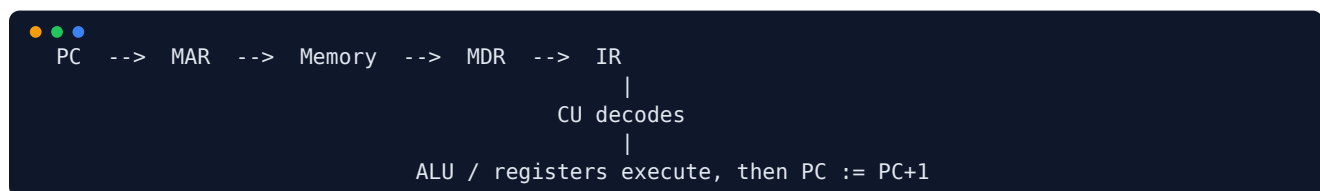


Fig. 3.1 Instruction cycle (simplified)

- Fetch: CU places PC on the address bus, issues READ, copies the instruction into IR, increments PC.
- Decode: CU interprets the opcode and turns on the correct control lines.
- Execute: ALU or a data movement happens; result may go to a register or memory.
- Store / write-back (sometimes listed separately): result is written back if needed.

Q. What is the role of the CPU? Name its two main parts.

A. The CPU processes instructions. Its two main parts are the ALU (arithmetic/logic) and the Control Unit (sequencing). Registers and cache support both.

Unit 4 — Operating System

Curriculum weight 10%. 'What is an OS and why is it necessary?' is a standard 3-mark or 8-mark question.

4.1 Definition and objectives

Definition — Operating system

System software that manages hardware resources and provides a platform on which application programs run. It is the first program the user effectively meets after firmware/BIOS.

Objectives: convenience for the user, efficient use of CPU/memory/devices, security, and a stable environment.

Common desktop/server families: Microsoft Windows, Apple macOS, UNIX/Linux, and historically MS-DOS. Mobile: Android, iOS.

4.2 Features / types

Type	Idea	Example
Batch	Jobs queued; no interaction while running	Old payroll runs

Type	Idea	Example
Multi-tasking	Several programs appear to run at once (time slices)	Browser + Word
Time-sharing	Many users, short CPU bursts each	University server
Multi-processing	More than one CPU/core sharing work	Quad-core PC
Parallel	Problem split across processors at once	Scientific jobs
Distributed	Many machines act as one resource pool	Cluster
Embedded	OS inside a device, limited UI	Car ECU, router
Real-time	Deadlines must be met	Airbag controller

4.3 Functions of an OS

- Booting — loading the kernel from disk into RAM after POST.
- Process management — create, schedule, suspend, terminate processes.
- Memory management — allocate/free RAM, virtual memory / paging.
- File management — names, folders, permissions, FAT/NTFS/ext4.
- I/O and device management — drivers, buffers, spooling for printers.
- Secondary storage management — free space, disk scheduling.
- Network management — protocol stack, sharing.
- Protection and security — passwords, access control, user accounts.
- Command interpreter / shell — CMD, PowerShell, bash, or a GUI.

4.4 Process management

Definition — Process

A program in execution. A program on disk is passive; a process has a program counter, registers, and an address space.

State	Meaning
New	Process is being created
Ready	Waiting for CPU; all other resources available
Running	Instructions are executing on a CPU
Waiting / blocked	Waiting for I/O or an event
Terminated	Finished or killed; PCB will be freed

Thread: a lightweight path of execution inside a process. Multi-threading = many threads in one process (a browser with one tab hanging). Multi-tasking = many processes (browser + Word).

4.5 Working with a GUI OS (Windows skills)

- Files and folders: create, delete, copy, rename, drag-and-drop.
- Path: C:\Users\Ali\Documents\notes.pdf — drive, folders, file.
- Search by name, date, or type; wildcards * and ?.
- Attributes: read-only, hidden, system, archive.
- Control Panel / Settings: printers, accounts, date/time, programs.
- dxdiag — DirectX diagnostic of display/sound/CPU.
- msconfig — startup and boot options (use carefully).

4.6 Open source and mobile OS

Open-source OS (Linux distributions) publish source code; users may study, change, and share under a licence. Advantages: low cost, community patches, control, good servers. Android is a mobile OS based on a Linux kernel, versioned as API levels (mention any current name you know, e.g. recent Android releases).

Q. Why is an operating system necessary?

A. Without an OS, every program would have to talk to hardware itself. The OS hides hardware details, shares the CPU fairly, keeps files organised, and stops one program from destroying another's memory.

Extra A — Number systems and codes

Not a separate 2019 unit, but BIEK Paper-I still examines conversions and character codes (ASCII, BCD, EBCDIC). Learn the methods; examiners love a conversion with working.

A.1 Four number systems

System	Base	Digits	Board use
Decimal	10	0-9	Human counting
Binary	2	0, 1	Machine states
Octal	8	0-7	Compact binary (groups of 3)
Hexadecimal	16	0-9, A-F	Compact binary (groups of 4); colours, addresses

A.2 Conversion methods

- Decimal → binary: divide by 2, remainder is the next low-order bit; read remainders bottom-up.
- Binary → decimal: expand weights 1, 2, 4, 8, 16, ... from the right.
- Binary ↔ octal: groups of 3 bits. Binary ↔ hex: groups of 4 bits. Pad with leading zeros.
- Decimal → hex: divide by 16; remainders 10-15 become A-F.

Worked example

Convert 45 to binary: $45/2=22$ r1, $22/2=11$ r0, $11/2=5$ r1, $5/2=2$ r1, $2/2=1$ r0, $1/2=0$ r1. Read up: 101101_2 .

Check: $32+8+4+1=45$. Hex: $101101 \rightarrow 0010\ 1101 = 2D_{16}$.

A.3 Character codes

Code	Bits / character	Note
BCD	4 bits per decimal digit	8421 weights; $45 \rightarrow 0100\ 0101$
EBCDIC	8 bits	IBM mainframes
ASCII	7-bit original, 8-bit extended	A = 65, a = 97, 0 = 48, space = 32
Unicode / UTF-8	variable	All world scripts; mention as modern extra

Board exam tip

If an MCQ says 'standard code for microcomputers', the answer is almost always ASCII.

Extra B — Boolean algebra and logic gates

A high-frequency BIEK Paper-I topic. Practise truth tables until you can draw them in under a minute.

B.1 Why Boolean algebra?

Definition — Boolean algebra

Algebra of two values, TRUE/FALSE or 1/0, using AND, OR and NOT. It is the mathematics of digital circuits and of conditions in programs.

AND ($\cdot$) is 1 only if every input is 1. OR ($+$) is 1 if any input is 1. NOT ($'$) inverts a single input.

B.2 Basic gates

Gate	Expression	Output 1 when	Unique identity
AND	$Y = A \cdot B$	all inputs 1	
OR	$Y = A + B$	at least one input 1	
NOT	$Y = A'$	input is 0	one input only
NAND	$Y = (A \cdot B)'$	not (all 1)	universal gate
NOR	$Y = (A + B)'$	all inputs 0	universal gate
XOR	$Y = A \oplus B$	inputs differ	odd number of 1s
XNOR	$Y = (A \oplus B)'$	inputs same	equality detector

Remember

NAND and NOR are universal: any Boolean function can be built from only NAND gates, or only NOR gates. That fact is a popular MCQ.

B.3 Truth tables of AND, OR, NOT, NAND

A	B	AND	OR	NAND	NOR	XOR
0	0	0	0	1	1	0
0	1	0	1	1	0	1
1	0	0	1	1	0	1
1	1	1	1	0	0	0

B.4 Laws (learn names + one identity each)

Law	AND form	OR form
Identity	$A \cdot 1 = A$	$A + 0 = A$
Null / Annulment	$A \cdot 0 = 0$	$A + 1 = 1$
Idempotent	$A \cdot A = A$	$A + A = A$
Complement	$A \cdot A' = 0$	$A + A' = 1$
Commutative	$A \cdot B = B \cdot A$	$A + B = B + A$
Associative	$A \cdot (B \cdot C) = (A \cdot B) \cdot C$	$A + (B + C) = (A + B) + C$
Distributive	$A \cdot (B + C) = A \cdot B + A \cdot C$	$A + (B \cdot C) = (A + B) \cdot (A + C)$

Law	AND form	OR form
Absorption	$A \cdot (A+B) = A$	$A + (A \cdot B) = A$
Involution	$(A')' = A$	$(A')' = A$
De Morgan	$(A \cdot B)' = A' + B'$	$(A + B)' = A' \cdot B'$

Principle of duality: swap $\cdot$ with $+$ and 0 with 1; the dual identity is also true.

B.5 De Morgan — the 8-mark favourite

Law 1: $(A \cdot B)' = A' + B'$. Law 2: $(A + B)' = A' \cdot B'$. Prove each with a truth table AND a gate diagram.

A	B	A · B	(A · B)'	A'	B'	A' + B'
0	0	0	1	1	1	1
0	1	0	1	1	0	1
1	0	0	1	0	1	1
1	1	1	0	0	0	0

Columns $(A \cdot B)'$ and $A' + B'$ match, so Law 1 is proved. Repeat the pattern for Law 2 in the exam.

B.6 SOP, POS, and Karnaugh map (idea)

From a truth table, a 1-row becomes a minterm (AND of literals). Sum of those minterms is SOP. A 0-row becomes a maxterm (OR of literals); product of maxterms is POS. A K-map groups 1s in power-of-two blocks (1, 2, 4, 8) to cancel variables. BIEK usually stays at 2 or 3 variables; FBISE-style 4-variable maps are extra credit.

Example: $Y = AB + BC$ reduces with Boolean algebra; $Y = CD + EF$ needs two AND gates and one OR gate.

Extra C — Data security

Definition — Malware

Software written to damage, steal, or hijack a system. A virus attaches to files and spreads when they run; a worm self-replicates over networks; a Trojan looks useful but hides a payload; ransomware encrypts files for money.

- Antivirus / anti-malware scans signatures and behaviour; update definitions daily.
- Firewall filters packets by rule (port, IP, application); can be hardware or software.
- Proxy server sits between clients and the Internet: cache, filter, hide internal IPs.
- Email threats: phishing, fake invoices, attachments with macros. Never send passwords by email.
- Good habits: strong unique passwords, backups, licensed software, least privilege accounts.

Q. Why do we use antivirus software? Name two products.

A. To detect, quarantine, and remove malware before it damages files or steals data. Examples: Windows Defender, Kaspersky, Avast, Bitdefender (any two named correctly).

Unit 5 — Programming concepts using C++

Curriculum weight 20% — the largest XI theory unit. BIEK Paper-II Option I still uses C (printf/scanf). The 2019 XI book uses C++ streams. Learn both syntaxes; the ideas (types, if, loops, functions) are the same.

5.1 What is a program?

Definition — Program

A finite set of instructions written in a programming language to solve a problem.

Language levels: machine (binary), assembly (mnemonics + assembler), high-level (C, C++, Java, Python). Translators: compiler (whole program → object code), interpreter (line by line), assembler (assembly → machine). A linker combines object files; an IDE (Turbo C++, Dev-C++, Code::Blocks, VS Code) edits, compiles, and runs.

Error type	When it is found	Example
Syntax	Compile time	Missing semicolon
Semantic / type	Compile or link time	Assigning a string to int wrongly
Runtime	While running	Divide by zero, missing file
Logical	Wrong answer, program 'works'	Using = instead of == in a condition

5.2 Structure of a C++ program

```
#include <iostream>
using namespace std;

int main() {
    int a, b, sum;
    cout << "Enter two integers: ";
    cin >> a >> b;          // cascading input
    sum = a + b;
    cout << "Sum = " << sum << endl;
    return 0;
}
```

Program 5.1 IPO with cascading streams

Turbo C style of the same idea uses #include <stdio.h>, printf/scanf, and often void main() plus getch(). BIEK Option I papers still show that style in Class XII.

5.3 Tokens, data types, operators

- Tokens: keywords (int, if, return), identifiers (marks1 — must start with letter or _), constants, operators, punctuators, strings.
- Basic types: int, short, long, float, double, char, bool (C++), void.
- Arithmetic: + − * / % Relational: < <= > >= == != Logical: && || ! Assignment: = += -= *= /= %=
- Increment: ++x (pre) vs x++ (post). Conditional: c ? t : f. sizeof operator.

Board exam tip

Precedence: postfix ++, then unary, then * / %, then + −, then relational, then ==, then &&, then ||, then assignment. When in doubt, use parentheses. $y = 5 + 3 - 6 * 2 / 3$ does * and / first: $6 * 2 = 12$, $12 / 3 = 4$, then $5 + 3 - 4 = 4$.

Common mistake

`a+=1` is NOT increment. Valid increment forms: `a = a+1`; `a += 1`; `a++`; `++a`.

5.4 Selection

```

if (marks >= 40)
    cout << "Pass";
else
    cout << "Fail";

// nested / else-if ladder
if (n > 0) cout << "positive";
else if (n < 0) cout << "negative";
else cout << "zero";

switch (colour) {
    case 1: cout << "Red"; break;
    case 2: cout << "Black"; break;
    case 3: cout << "White"; break;
    default: cout << "No colour";
}

```

if, else-if, and switch

Remember

switch can test int or char, not float. Always write break unless you want fall-through.

5.5 Iteration

Loop	Test	Guaranteed to run once?
for (init; test; update)	before body	No (if test is false at start)
while (test)	before body	No
do { body } while (test);	after body	Yes

```

// print 1 to 5
for (int i = 1; i <= 5; i++)
    cout << i << " ";

int i = 1;
while (i <= 5) { cout << i << " "; i++; }

i = 1;
do { cout << i << " "; i++; } while (i <= 5);

```

Three loops, same output

Nested loops build tables and patterns. break leaves the loop; continue skips to the next iteration; goto is legal but poor style.

5.6 Functions

A function is a named block. Advantages: reuse, shorter main, easier testing. Predefined: sqrt, pow (cmath), strlen (cstring). User-defined: you write them.

- Signature = name + parameter types + return type.
- Prototype (declaration) tells the compiler before main; definition has the body; call uses the name.
- Formal parameters are in the definition; actual parameters are in the call.
- Pass by value copies; pass by reference (int &x) can change the caller's variable.

- Default arguments: `void greet(string n = "Student");`
- local vs global vs static: local dies when the function ends; global lives for the program; static local remembers its value between calls.
- `inline` suggests the compiler paste the body at the call site (small functions).
- `exit(0)` (`cstdlib`) ends the whole program.

```

#include <iostream>
using namespace std;

int larger(int x, int y);           // prototype

int main() {
    cout << larger(12, 7);
    return 0;
}

int larger(int x, int y) {          // definition; x,y formal
    return (x > y) ? x : y;
}

```

Prototype, call, definition

5.7 Function overloading

Same function name, different parameter lists (count or types). Return type alone is not enough to overload. The compiler picks the best match at compile time (static polymorphism).

```

int add(int a, int b) { return a + b; }
double add(double a, double b) { return a + b; }
int add(int a, int b, int c) { return a + b + c; }

```

Unit 6 — Arrays, strings and structures

Curriculum weight 15%. Sorting, searching, and string reverse appear in both theory and practicals.

6.1 One-dimensional arrays

Definition — Array

A named sequence of elements of the same type stored in consecutive memory locations, accessed by an integer index starting at 0.

```

int marks[5] = {70, 80, 90, 60, 75};
cout << marks[0];           // 70
int n = sizeof(marks) / sizeof(marks[0]); // 5

int sum = 0;
for (int i = 0; i < 5; i++)
    sum += marks[i];
cout << "Average = " << sum / 5.0;

```

Declaration, index, traversal, sizeof

Common mistake

`marks[5]` on an array of size 5 is out of range (valid indices 0–4). Board MCQs love this trap.

6.2 Two-dimensional arrays

Think of a matrix: rows then columns. `int a[3][3];` Access `a[r][c]`. Nested loops required.

```

int a[2][2] = {{1, 2}, {3, 4}};
int b[2][2] = {{5, 6}, {7, 8}};
int c[2][2];
for (int i = 0; i < 2; i++)
    for (int j = 0; j < 2; j++)
        c[i][j] = a[i][j] + b[i][j];

```

Matrix addition (same pattern for 3×3)

6.3 Strings

A C-string is a char array ending with '\0'. char name[20] = "Ali"; occupies A,l,i,\0. Never forget the terminator when counting storage.

Function	Header	Job
strlen(s)	<cstring>	Count characters before \0
strcpy(d,s)	<cstring>	Copy s into d
strcat(d,s)	<cstring>	Append s onto d
strcmp(a,b)	<cstring>	0 if equal, <0 if a<b, >0 if a>b
substr (C++ string)	<string>	s.substr(pos, len)

```

#include <iostream>
#include <cstring>
using namespace std;
int main() {
    char name[30];
    cin.getline(name, 30);
    for (int i = strlen(name) - 1; i >= 0; i--)
        cout << name[i];
    return 0;
}

```

Print a name in reverse using strlen

6.4 Structures

Definition — Structure

A user-defined type that groups variables of possibly different types under one name. Unlike an array, members can mix char, int, float.

```

struct Employee {
    char name[30];
    char designation[20];
    float salary;
};

int main() {
    Employee e = {"Sara", "Clerk", 45000};
    cout << e.name << " " << e.designation << " " << e.salary;
}

```

Curriculum practical: display employee data

Dot operator e.name accesses a member. An array of struct Employee staff[50]; is how simple databases start — a bridge to Class XII.

Unit 7 — Computer communication and networks

Curriculum weight 15%. Topologies with diagrams are a classic Section C question.

7.1 Network and components

Definition — Computer network

Two or more computers connected to share resources (files, printers, Internet) and to communicate.

Component	Role
Sender	Device that originates the message
Receiver	Device that accepts the message
Message	Data actually sent (file, voice, video)
Medium	Path: cable or wireless
Protocol	Agreed rules (HTTP, TCP, IP, SMTP)

Advantages: sharing, communication, cheaper peripherals, central backup. Disadvantages: security risk, cabling cost, dependence on the server in a star.

7.2 Communication modes (data flow)

Mode	Direction	Example
Simplex	One way only	Keyboard → CPU, radio broadcast
Half duplex	Both ways, not at the same time	Walkie-talkie
Full duplex	Both ways at once	Telephone, modern Ethernet

7.3 Transmission media

Kind	Examples	Notes
Guided (wired)	Twisted pair, coaxial, fibre optic	Fibre: light, longest distance, immune to EMI
Unguided (wireless)	Radio, microwave, infrared	Microwave often needs line of sight

Twisted pair: cheap LAN cable (Cat5e/Cat6). Coaxial: older TV and some backbone. Straight-through cable: PC to switch. Crossover: PC to PC (older NICs). Practicals ask you to crimp RJ-45.

7.4 Network types

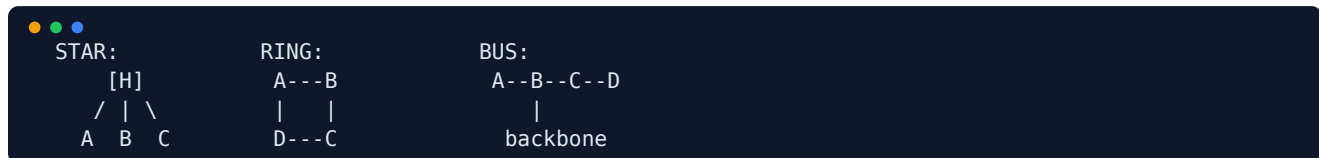
Type	Span	Example
LAN	Room, floor, campus	College computer lab
MAN	A city	City-wide cable network
WAN	Country / planet	The Internet
PAN (extra)	A few metres	Phone + earbuds (Bluetooth)

Definition — Internet / Intranet

Internet is the global public network of networks using TCP/IP. Intranet is a private TCP/IP network of one organisation, often using web technology but not open to the world.

7.5 Topologies

Topology	Shape	Strength	Weakness
Bus	One backbone	Cheap cable	Backbone break kills all
Star	All nodes to a hub/switch	Easy to add PCs; one cable fault is local	Central device is a single point of failure
Ring	Closed loop	Fair access (token)	A break can stop the ring (unless dual ring)
Mesh	Many-to-many links	Very fault tolerant	Expensive cabling
Tree	Hierarchy of stars	Scales a campus	Root failure
Hybrid	Mix of the above	Fits real buildings	Complex to manage



Sketch these three in Section C; label hub/switch on star

7.6 Standards and architectures

Body	What it standardises
ISO	OSI 7-layer model; many quality standards
IEEE	LAN standards, e.g. 802.3 Ethernet, 802.11 Wi-Fi
ITU	Telecom (radio, telephone, some wireless)
ASCII (as a code standard)	Character encoding — often listed beside these bodies in the curriculum

ISO OSI seven layers (learn in order)

#	Layer	Job in one line
7	Application	User programs: HTTP, SMTP, FTP, DNS
6	Presentation	Encryption, compression, format (JPEG, ASCII)
5	Session	Start, manage, end conversations
4	Transport	End-to-end: TCP (reliable), UDP (fast)
3	Network	Routing and IP addressing
2	Data link	Frames, MAC addresses, switches
1	Physical	Bits on wire/radio, cables, hubs

TCP/IP model (Internet model) is the one the Internet actually uses: Application, Transport, Internet, Network Access (Link). Mnemonic for OSI: All People Seem To Need Data Processing.

Q. What is topology? Describe star, bus and ring with diagrams.

A. Topology is the geometric arrangement of links and nodes. Draw the three sketches, then write one advantage and one disadvantage for each (use the table above).

Extra D — Signals, modem and software types

Old BIEK Paper-I still asks analog versus digital signals, the job of a modem, and a full software classification. Keep this extra lecture next to Unit 7.

D.1 Analog and digital signals

Point	Analog	Digital
Form	Continuous wave, infinite values in a range	Discrete levels, usually two (0 and 1)
Example	Human voice on a copper telephone line, mercury thermometer	CD audio after sampling, a USB transfer
Noise	Hard to separate from the signal	Easier to regenerate; computers are digital
Use in PCs	Sound card input before conversion	CPU, RAM, storage, Ethernet

Definition — Modem

Modulator–demodulator. It converts a computer’s digital bits into analog waves for a telephone or cable line (modulation) and converts incoming analog waves back into bits (demodulation). ADSL/fibre boxes at home still play this role on the last hop.

D.2 Full software map (8-mark structure)

Definition — Software

A set of programs, procedures and related documentation that tell hardware what to do. Without software a computer is a dead box.

Branch	Sub-type	Examples
System software	Operating system	Windows, Linux, Android, macOS
System software	Device driver	Printer driver, graphics driver
System software	Language translator	Compiler (gcc), interpreter (Python), assembler
System software	Anti-malware	Windows Defender, Kaspersky
Application — general	Office, browser, media	Word, Excel, Chrome, VLC
Application — special	One organisation / job	Hospital MIS, bank CBS, college payroll
Utility	Housekeeping	Zip, backup, Disk Cleanup, Recuva

Translators in more detail: a compiler reads the whole source and builds an object/exe (fast later runs, all syntax errors at once). An interpreter executes line by line (easier to try, slower, stops at the first runtime error). An assembler translates mnemonic assembly into machine code, one instruction roughly to one opcode.

Q. Define software. Explain system software and application software.

A. Start with the definition box. Draw the classification table. Give two examples in each box. Close with: system software manages the machine; application software does the user’s actual job (letter, accounts, game).

Grade XI practical journal (Sindh list)

25 marks. Keep a dated journal. The official activity list includes hardware, Windows, C++, and networking.

Hardware and OS

- Identify CPU, RAM, HDD/SSD, PSU, ports on a motherboard.
- Assemble/disassemble RAM, drive, and SATA/power cables (under teacher supervision).
- BIOS/UEFI: boot order, date/time, hardware monitor.
- Desktop, Explorer, create/copy/move/delete/rename, search, attributes.
- Control Panel resource management; dxdiag; msconfig (view only if school policy says so).

Programming (write these in your journal with output)

- 1 Calculator using arithmetic operators.
- 2 Mark sheet with if / else-if (grade A/B/C/F).
- 3 Prime or composite test.
- 4 Split a 4-digit number onto four lines.
- 5 Factors of n; stars pattern function; larger-of-two function.
- 6 Sort an array; linear search; average of array; 3×3 add/multiply.
- 7 Reverse a name with strlen; employee structure display.

Networking practicals

- Identify CAT cable, crimp straight-through and crossover, test with a tester.
- Name switch, router, NIC, modem in the lab.
- Set a static IP, subnet, gateway, DNS; ping the gateway.
- Share a folder on a LAN; explain client vs centralised vs workgroup.

Model practice paper — Computer Science I

Time 3 hours. Marks 75. Attempt this on paper, then mark from the keys below.

Section A — MCQs (attempt all)

1. The brain of the computer is the

- (A) ALU only
- (B) CPU ✓
- (C) Monitor
- (D) USB

2. 1 MB equals

- (A) 1000 KB
- (B) 1024 KB ✓
- (C) 1024 bytes
- (D) 1024 GB

3. Which memory is volatile?

- (A) ROM
- (B) Flash
- (C) RAM ✓
- (D) HDD

4. EEPROM is erased by

- (A) UV light
- (B) A hammer

- (C) Electric signal ✓
 - (D) BIOS password
5. The address bus is normally
- (A) Bidirectional
 - (B) Unidirectional ✓
 - (C) Optical
 - (D) Analog
6. Which is a function of an OS?
- (A) Compiling C++
 - (B) Process management ✓
 - (C) Crimping cable
 - (D) None
7. A process waiting for a printer is in
- (A) Running
 - (B) Ready
 - (C) Waiting/blocked ✓
 - (D) New
8. ASCII code of 'A' is
- (A) 48
 - (B) 65 ✓
 - (C) 97
 - (D) 32
9. $(A \cdot B)'$ equals
- (A) $A' \cdot B'$
 - (B) $A' + B'$ ✓
 - (C) $A + B$
 - (D) $A \cdot B$
10. A universal gate is
- (A) AND
 - (B) XOR
 - (C) NAND ✓
 - (D) Buffer
11. HTTP works at OSI layer
- (A) Transport
 - (B) Network
 - (C) Application ✓
 - (D) Physical
12. College lab network is typically a
- (A) WAN
 - (B) LAN ✓
 - (C) MAN
 - (D) GAN
13. Optical fibre carries
- (A) Sound
 - (B) Light ✓
 - (C) Water
 - (D) Steam
14. Compiler translates
- (A) Line by line
 - (B) The whole program ✓
 - (C) Only comments
 - (D) Only HTML
15. Index of the first array element is
- (A) 1
 - (B) 0 ✓

(C) -1

(D) n

16. Which loop always runs at least once?

(A) for

(B) while

(C) do-while ✓

(D) goto

17. strcmp returns 0 when strings are

(A) Unequal

(B) Equal ✓

(C) Empty

(D) Numeric

18. A firewall is used for

(A) Cooling CPU

(B) Filtering network traffic ✓

(C) Printing

(D) Compiling

19. Cloud computing mainly provides resources

(A) Only offline

(B) Over the Internet ✓

(C) Inside ROM

(D) Via plotter

20. HDMI is a

(A) Storage device

(B) Port ✓

(C) Topology

(D) Virus

21. Cache memory is usually

(A) DRAM

(B) SRAM ✓

(C) Magnetic

(D) Optical

22. Star topology's centre is often a

(A) Repeater only

(B) Hub or switch ✓

(C) Plotter

(D) Scanner

23. void as a return type means

(A) Returns int

(B) Returns nothing ✓

(C) Returns float

(D) Error

24. Which is application software?

(A) Linux kernel

(B) MS Word ✓

(C) Device driver

(D) BIOS

25. Full duplex example

(A) Radio

(B) Keyboard

(C) Telephone ✓

(D) Loudspeaker

26. Program Counter holds

(A) Current instruction

(B) Next instruction address ✓

(C) Result of ALU

(D) Stack size

27. 1 nibble is

(A) 2 bits

(B) 4 bits ✓

(C) 8 bits

(D) 16 bits

28. Phishing is usually delivered by

(A) Heatsink

(B) Email or fake websites ✓

(C) ALU

(D) HDMI

29. sizeof(int array of 10) / sizeof(int) gives

(A) 10 ✓

(B) 40

(C) 0

(D) 2

Section B — sample short questions (attempt 10 of 15)

- 1 Define computer system. List four characteristics.
- 2 Hard copy versus soft copy.
- 3 Volatile versus non-volatile memory with examples.
- 4 Differentiate compiler and interpreter.
- 5 Draw De Morgan's first law with a truth table.
- 6 LAN versus WAN.
- 7 Internet versus intranet.
- 8 Any three input devices and one use each.
- 9 Two types of printers.
- 10 What is a process? Name four states.
- 11 Define protocol and give two examples.
- 12 What is an array? Show a declaration.
- 13 Local versus global variables.
- 14 Purpose of BIOS and CMOS battery.
- 15 Two advantages of fibre optic cable.

Section C — sample detailed questions (attempt 2 of 3)

- 1 Define software. Explain system software and application software with examples.
- 2 What is topology? Explain star, bus and ring with diagrams, advantages and disadvantages.
- 3 Explain the fetch-decode-execute cycle with a labelled sketch of CPU components.

Quick glossary (XI)

Term	One-line meaning
ALU	Part of CPU that calculates and compares
BIOS/UEFI	Firmware that starts hardware before the OS
Bit	Binary digit 0 or 1
Bus	Shared communication pathway
Cache	Small fast memory close to the CPU

Term	One-line meaning
Compiler	Translator of a whole high-level program
GUI	Graphical user interface (windows, icons, mouse)
IDE	Program that edits, compiles and runs code
IPO	Input-Process-Output model
LAN	Local area network
OSI	ISO's seven-layer network model
Protocol	Rules of communication
RAM	Volatile main memory
ROM	Non-volatile firmware memory
SDLC	Covered in XII — lifecycle of software
USB	Universal Serial Bus port/protocol

End of Class XI lectures. Continue with the Class XII volume for C / Visual Basic options and the remaining 2019 units (SDLC, pointers, OOP, files, database, multimedia, wireless).